

Gesu India

HCI and Responsible AI researcher



Website



Email



Google Scholar



LinkedIn

Research Interests and Skills

I am a Computer scientist using mixed methods to design technologies for people with diverse abilities and socio-economic backgrounds. My specialities lie in Human-computer interaction, Responsible AI, and accessibility. My PhD was supported by the prestigious EPSRC Industrial CASE award with Microsoft Research, Cambridge. My supervisors were: Prof. Matt Jones, Prof. Jennifer Pearson, Prof. Simon Robinson and Dr. Cecily Morrison (MSR Boston). During my PhD, I designed a disability-first participatory approach to collecting AI dataset with low-resource communities. I have worked with multi-disciplinary teams and communities/stakeholders across geographies and cultures. My research work and interests span responsible AI, public health, education, usable security, and more. My work has resulted in commercially available products (JoJo Blocks and Braille Cards), a patent (TouchPIN), AI dataset (ORBIT-India), and numerous publications at top-tier HCI venues such as CHI, ASSETS, and ICTD.

Doctoral Research

PhD Researcher

Swansea University

2021 - 2025

- A mixed-methods study exploring the functional and social accessibility of object-recognition technologies for blind/low-vision individuals in India, home to the world's largest population of people with blindness. (*Paper here.*)
- Contextualised the high-resource Global North AI dataset collection tools and protocols to the resource-variant, socio-culturally diverse Indian context, creating the ORBIT-India dataset, the first teachable-object recognition dataset by blind/low-vision people in India. (*Paper here.*)
- A six-month situated ethnography-based research exploring the intricate intersection of disability and data work in India. This work unravels the lingering impacts of ghost work and history of social/systemic marginalisation entangled with community-spirit and personal motivations for accessing better AI systems, within the Indian context. [*Manuscript under review*]

Work Experience

Research Assistant

Swansea University Swansea, UK

Nov 2025-March 2026

- Explored the visions emergent users and technology stakeholders in India and Kenya have for the future of AI systems. [*Manuscript under review*]
- Studied the usage of residual energy from household equipments to energise and interact with a digital (IoT) memory harvesting system [*Manuscript under review*]
- Field testing of the in-house developed (open-source) tools for language technology design and development with tribal communities in North-East India [*Manuscript under review*]
- Designed lab studies to examine the relationship of human-generative AI collaborations for creative tasks.

Teaching Assistant

Swansea University Swansea, UK

Jan-April 2025

- Delivered numerous interactive workshops for course CS200 -Introduction to Human-Computer Interaction to a cohort of 100+ undergraduate students.
- Mentored students regarding troubleshooting project roadblocks and clarifying HCI concepts and frameworks.

- Managed the grading and assessment of weekly assignments and examinations.

Research Fellow

Microsoft Research *Bangalore, India*

2019-2021

- A nine-month ethnographic research addressing the socio-technical challenges of equitable deployment and localisation of computational thinking tools (Project Torino/Code Jumper) within the low-resource schools for blind/low-vision children in India, collaborating closely with local NGOs, educators, and other stakeholders to address unique accessibility and contextual challenges. (*Paper*)
- A mixed-methods study designing and evaluating an app that encourages walking habits among blind/low-vision users by enabling them to virtually explore the real world as they walk at home. (*Paper*)
- Contributed to the design and development of two numeracy toy kits for blind/low-vision children, called Jojo Blocks and Braille Numeracy Cards. Collaborations with multi-disciplinary teams, product vendors, and community stakeholders (educators, parents, and NGOs), also leading to the design, development, and testing of curriculum-aligned games to scaffold numeracy skill acquisition.
- Contributed foundational research for Microsoft's Project SEEDS), introducing digital games in schools for blind and low-vision children in India. Collaborated with the local education stakeholders to assess needs, preferences, and perceptions around integrating digital games into classroom environments. (*Paper*)
- An interview-based study exploring the lives of people with disabilities, focusing on the influence of interpersonal relationships, institutions, and assistive technologies on life aspirations and access to opportunities. This work has guided subsequent initiatives within Project SEEDS.
- Conducted early research studies applying and advancing the Ludic Design for Accessibility framework through multidisciplinary projects.

Research Intern

Microsoft Research *Bangalore, India*

July 2018-Jan 2019

- Designed and conducted initial research aiming to understand the role of vision, socio-economic, and educational background of children with mixed visual abilities in learning numeracy via examining number line estimation skills.
- Conducted multiple co-design workshops with educators of blind/low-vision children to understand existing numeracy teaching methods and create their game/play-based translations for classroom settings.

Research Intern

Singapore University of Technology and Design and IRIT (France) *Singapore* *May-June, 2017*

- Conducted a contextual inquiry to identify pain points and opportunities for supporting mobility instructors, further designing a multi-modal prototype (Android app paired with a tangible game) to support spatial skills for blind/low-vision teenagers in Singapore.

Research Intern

Indian Institute of Technology *Mumbai, India*

2016-17

- Developed a novel multi-modal smartphone authentication system for blind/low-vision smartphone users in India, as part of a multidisciplinary project spanning accessibility, usable security, and user-centered design.
- Two-phase user testing process assessing the system's accessibility and effectiveness in mitigating shoulder-surfing vulnerabilities in Indian public settings. The research formed the foundation for the granted European patent (EP3945402B1). (*Paper*)

Research Intern

Indian Institute of Technology *Guwahati India*

Dec, 2015

- Designed a novel board game to teach STEM concepts to rural Assam primary school students, fostering interactive and engaging learning experiences.

Real-World Impact

AI dataset

- *ORBIT-India dataset*: First teachable-object recognition dataset by blind/low-vision data collectors in India.

Commercial Products

- *JoJo Blocks*: Tactile blocks teaching numeracy to blind and low-vision children.
- *Braille Numeracy Cards*: Accessible Braille card deck for numeracy and inclusive play.

Patent Contribution

- *TouchPIN*: Accessible smartphone authentication system.

Peer-Reviewed Conference Publications

- CHI 2026 The ORBIT India Dataset: Understanding the Challenges of Collecting a Disability-First AI Dataset in Low-Resource Environments. **Gesu India**, Martin Grayson, Cecily Morrison, Daniela Massiceti, Simon Robinson, Jennifer Pearson, Matt Jones (Full-paper)
- CHI 2025 Exploring the Experiences of Individuals Who are Blind or Low-Vision Using Object-Recognition Technologies in India. **Gesu India**, Simon Robinson, Jennifer Pearson, Cecily Morrison, Matt Jones (Full-paper)
— Received 4-star rating in University internal REF mini-review
- ASSETS 2021 VStroll: An Audio-based Virtual Exploration to Encourage Walking among People with Vision Impairment. **Gesu India**, Mohit Jain, Pallav Karya, Nirmalendu Diwakar, Manohar Swaminathan (Full-paper)
- INTERACT 2021 Understanding Motivations and Barriers to Exercise among People with Blindness in India. **Gesu India**, Mohit Jain, Manohar Swaminathan. (Full-paper)
- CHI 2021 Teachers' Perceptions around Digital Games for Children in Low-resource Schools for the Blind. **Gesu India**, Vidhya Y., Aishwarya O., Nirmalendu Diwakar, Mohit Jain, Aditya Vashistha, Manohar Swaminathan. (Full-paper)
- ICTD 2020 Conceptual Learning through Accessible Play: Project Torino and Computational Thinking for Blind Children in India. **Gesu India**, Geeta Ramakrishna, Joyojeet Pal, Manohar Swaminathan. (Full-paper)
- ASSETS 2019 Computational Thinking as Play: Experiences of Children who are Blind or Low Vision in India*. **Gesu India**, Geeta Ramakrishna, Jyoti Bisht, Manohar Swaminathan. (Full-paper)
*Recipient of Best Artifact Award (Runner Up)

Software & Skills

- SKILLS Qualitative research methods (interviews, focus groups, co-design workshops, diary study, and more) and Quantitative research methods (surveys, statistical analysis, A/B testing, and more), Ethnography, Product design, Prototyping (Low/high-fidelity), User Experience, User Interaction, Usability Testing
- SOFTWARE MAXQDA, Android Studio, MATLAB, Figma, InDesign, Photoshop, Illustrator, Miro, Adobe Creative Suite, Jupyter Notebook
- PROGRAMMING Python, JAVA, Web (HTML, CSS, JavaScript)

Education

Doctor of Philosophy, Computer Science

2021-2025

Swansea University, UK

Thesis Title: “Participatory AI Dataset Creation in the Global South”

Advisors: (Swansea University) Dr. Matt Jones, Dr. Simon Robinson, and Dr. Jennifer Pearson, and (Microsoft Research, Boston) Dr. Cecily Morrison

Bachelor of Technology, Civil Engineering

2014-2018

Indian Institute of Technology, Patna

Awards and Scholarships

UKRI 2021 EPSRC Industrial-CASE Award

MICROSOFT 2021 Microsoft Research PhD Fellowship

PRE-CHI 2026 Best Talk Award at the South West UK Pre-CHI event at Bristol University

ASSETS 2022 SIGACCESS Diversity and Inclusion Scholarship

ASSETS 2019 Artifact Award (Runner-Up)

INDIAHCI 2016 Best Poster Award

Workshop Articles and Whitepapers

ASSETS 2023 Bridging the Gap: Towards Advancing Privacy and Accessibility among People with Vision Impairment. *Rahaf Alharbi, Robin N Brewer, Gesu India, Lotus Zhang, Leah Findlater, Yixin Zou, Abigale Stangl. (Workshop Article)*

MICROSOFT 2021 Reimagining Accessibility and inclusion in K-12 CS Education through curriculum and professional development. *(Whitepaper)*

INTERACT 2017 TouchPIN: Numerical Passwords You Can Feel. **Gesu India.** *(Adjunct Proceedings)*

Panel, Presentations, and & Invited Talks

2026 (Panelist) “*The Future of AI*”. CDT Retreat 2026, Gregynog. Swansea University, UK.

CHI 2023 (Workshop) “*Behind the Scenes of Automation: Ghostly Care-Work, Maintenance, and Interferences*”

2022 (Panelist) “*Disability and Access to Public Digital Health Services*”. Digital Futures Lab, India. Recorded Video

EMPOWER 2019 (Presentation) “*Music, Storytelling and Play: Teaching Computational Thinking to Blind Children in India*”.

EPISTEME 2019 (Invited Talk) “*Ludic Design for Accessibility*”. Infosys, India

IIT BOMBAY 2019 (Invited Talk) “*Accessible Authentication for VI Smartphone User*”

INDIAHCI 2016 (Presentation) “*Haptics for Authentication*”

Leadership, Outreach & Services

- Technical Program Committee — ASSETS 2026 and HAXD 2026
- Reviewer — CHI (2023-present), FAccT (2024, 2026), ASSETS (2023), DIS (2026), CSCW (2024), TEI (2024)
- Student Volunteer — ASSETS (2022), INTERACT (2017), and IndiaHCI (2016)
- Organizer (Panel, 2023) for “*Charting an Ethical and Inclusive Path: Navigating the Future of AI and Data-Intensive Systems*” — Swansea University